

Chapter 198

Remote Grading and Remote Monitor

This chapter describes how to set up Remote Grading using two separate DaVinci Resolve systems in different locations via the internet, to have one system remotely control the other for color grading. While DaVinci Remote Monitor allows you to stream a high quality video signal from one DaVinci Resolve system to another workstation over the internet or on the same network.

Contents

Introduction to Remote Grading	4194
Requirements for Remote Grading	4194
Setting Up for Remote Grading	4194
Remote Grading Restrictions	4195
DaVinci Remote Monitor (Studio Version Only)	4195
Requirements for DaVinci Remote Monitor	4196
Setting Up DaVinci Remote Monitor	4196
DaVinci Remote Monitoring Using IP Address Connections	4199
DaVinci Remote Monitoring via User Specified TURN Servers	4200
DaVinci Remote Monitor Restrictions	4201

Introduction to Remote Grading

To enable colorists to work interactively with clients across the globe, DaVinci Resolve offers a remote grading option. It allows two matching DaVinci Resolve systems to be synchronized via an Internet connection such that changes made on the colorist's workstation are immediately applied on the remote client's workstation.

Cue commands are also synchronized to ensure that both systems are always on the same frame in the Timeline. Starting or stopping playback on the colorist's DaVinci Resolve also starts and stops the remote client system. While a remote grading session is in progress, input from the user at the remote client's DaVinci Resolve workstation is ignored.

Currently, the remote grading feature supports only color correction and does not allow editing or conforming during a session. The two colorist and remote DaVinci Resolve systems must have matching timelines and the number of clips, clip durations, and system resolutions must match. The requirements and limitations of a remote grading session are summarized below.

Requirements for Remote Grading

- The same version of DaVinci Resolve must be installed on both systems.
- The Timeline to be graded must be conformed on both machines prior to the start of the remote grading session.
- The number of clips on the Timeline and the duration of each clip must be identical.
- While grading, the active Timeline and versions on the remote client system are constantly updated. Creating, deleting, or switching the Timeline on the client's DaVinci Resolve is not allowed. Doing so will terminate the remote grading session immediately.
- You cannot make any grading adjustments on the remote client's DaVinci Resolve workstation until the remote grading session has ended.

NOTE: Remote grading does not require a shared project library.

Setting Up for Remote Grading

To start a remote grading session, the client's DaVinci Resolve must be able to connect to the colorist's system using TCP/IP.

- 1 Open DaVinci Resolve on the remote client's workstation (the one that's being remotely controlled), log in, and open the project that will be remotely graded.
- 2 Choose Workspace > Remote Grading (Ctrl-G) on the remote client's workstation. A window is displayed with text fields to enter the IP address and port number of the colorist's system.
- 3 Set the IP address field to the IP of the colorist's DaVinci Resolve workstation. If the colorist's system already has a public IP address, the port number can be left at its default value (15000). If the colorist's system is on a private network, the colorist or their network administrator should set the port number to one on the public IP router that is internally routed to port 15000 of the colorist's DaVinci Resolve.

- 4 Once the remote client clicks Connect, the client's DaVinci Resolve system will attempt to establish a connection with the remote colorist's workstation.
- 5 Once the connection is established, a pop-up appears on the colorist's screen asking for permission to accept a Remote Grading connection.
- 6 Click OK to accept, minimize the size of this dialog window, and continue grading normally.

The Remote Grading session will remain active until one of the users chooses to disconnect or an error occurs causing DaVinci Resolve to automatically terminate the session.

Remote Grading Restrictions

To allow operation over low bandwidth and a potentially long latency Internet connection there are some restrictions to remote operation.

- When playback is started, the playback speeds on the two DaVinci Resolve systems may differ. The frame positions are only guaranteed to be synchronized when playback is stopped.
- Input/output/display LUTs applied from the Config page on the colorist's DaVinci Resolve will not have any effect on the client system. LUTs selected on the client's DaVinci Resolve will be applied instead.
- Presets applied from the Config/Color pages on the colorist's system will not have any effect on the client's system. Presets selected on the client's DaVinci Resolve will be applied instead.

DaVinci Remote Monitor (Studio Version Only)

DaVinci Remote Monitor is an application that allows you to have access to a low latency, high quality video signal over a network for monitoring, editing, and color grading purposes. The DaVinci Remote Monitor shows the output of the host's Viewer in real time as they work in DaVinci Resolve. This allows producers to monitor a session, while editors and colorists have the ability to work remotely using the DaVinci Resolve interface and a data stream from a remote DaVinci Resolve workstation. This data stream is of high enough quality that you can run the signal through a Blackmagic DeckLink or UltraStudio device to a grading monitor and have the same confidence in the output as if it was connected locally.

The stream quality can be adjusted for your particular needs and available bandwidth. For example, an editor may only need an HD h.264 8-bit 4:2:0 codec for offline editing, while a colorist may need the full UHD h.265 12 bit 4:4:4 RGB codec for HDR grading.

NOTE: For clarity in this section we will refer to the main DaVinci Resolve workstation in the post house or data center that is streaming the video as the "Resolve Host" and the devices that are receiving the stream as the "Resolve Clients." However, in reality these are all just devices running the same version of the DaVinci Resolve Studio and the DaVinci Remote Monitoring apps.

Requirements for DaVinci Remote Monitor

In order to reliably stream a high-quality video signal from one system to another, there are some technical hardware and software limitations and requirements for the Host and Clients.

The Resolve Host must have the following hardware and software requirements for DaVinci Remote Monitor:

- The Resolve Host needs to have the Mac, Linux, or Windows version of DaVinci Resolve Studio installed.
- For Linux and Windows users, the Resolve Host needs an RTX series NVIDIA GPU and drivers installed. AMD and Intel GPUs are currently unsupported.
- The Host must have a Blackmagic Cloud account.

The Resolve Client must have the following hardware and software requirements for DaVinci Remote Monitor:

- The Resolve Client needs to have the Mac, Linux, or Windows version of DaVinci Resolve Studio installed. The DaVinci Remote Monitor App is automatically installed in the same folder as DaVinci Resolve.
- Apple iPhone and iPad devices are supported as Client platforms. Download the DaVinci Remote Monitor app from the App Store (The Studio Version of DaVinci Resolve is not required on these devices).
- For Linux and Windows users, the Resolve Client needs an RTX series NVIDIA GPU and drivers installed. AMD and Intel GPUs are currently unsupported.
- All Clients must have a Blackmagic Cloud account.

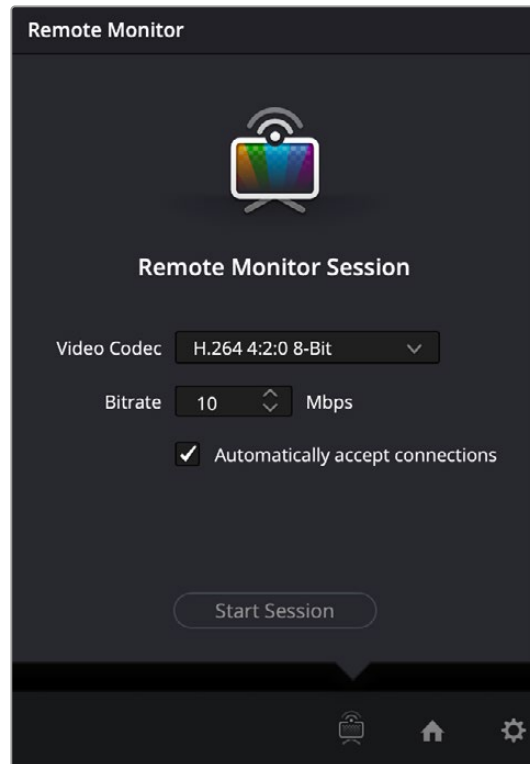
Setting Up DaVinci Remote Monitor

Setting up a Remote Monitor session is easily done from the DaVinci Resolve Studio interface: additionally, the Host and all Clients must have Blackmagic Cloud accounts.

To Start a DaVinci Remote Monitor session as the Resolve Host:

- 1 The Resolve Host needs to be running the full DaVinci Resolve Studio version of the software.
- 2 Select Workspace > Remote Monitor.
- 3 Sign into your Blackmagic Cloud account, if necessary.
- 4 In the Remote Monitor Window, select the Video Codec and Bitrate you wish to use for the session. Please note, the Video Codec used must be one that all the client devices support.

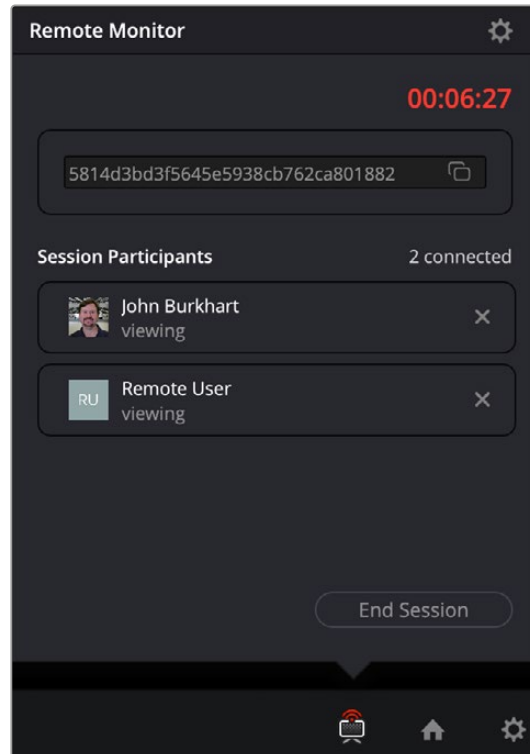
For maximum compatibility, H.264 4:2:0 8-bit is a good choice, however accurate color monitoring may necessitate using another 10- or 12-bit codec instead. Bandwidth needs to increase with the number of clients that are connected. While Remote Monitoring over the internet is possible, the best performance will be had with a wired ethernet connection.



The DaVinci Remote Monitor Session Setup

- 5 Check the “Automatically accept connections” box to let anyone with the Session Code connect directly. If you wish to approve each connection manually, uncheck this box.
- 6 Click the Start Session button.
- 7 Copy the generated Session Code from its field, and disseminate it to all the Resolve Clients via email, text message, etc.

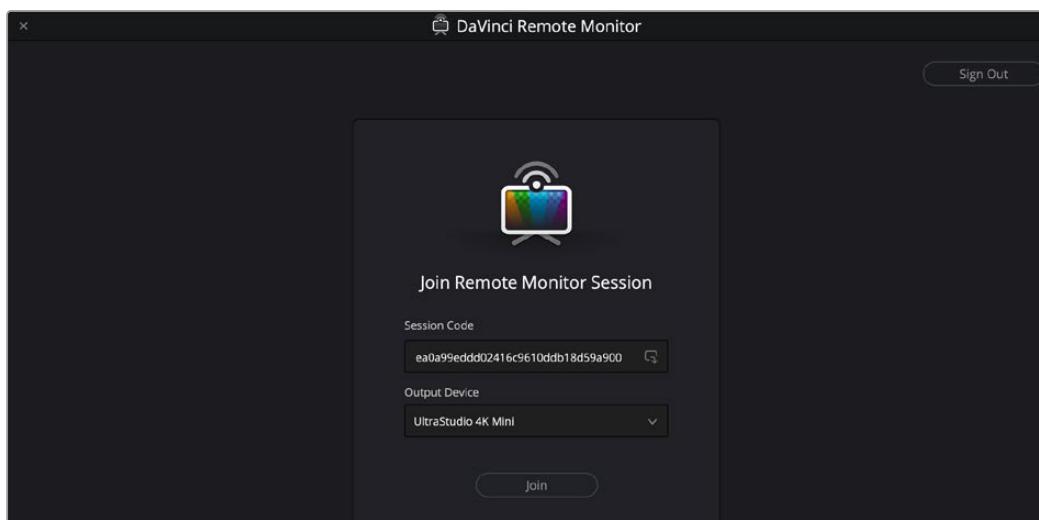
At this point your Remote Monitor has started, and a timer will show the duration of the current session. A Remote Monitor icon will appear in the lower right of the DaVinci Resolve interface, and the top of it will turn red to indicate that at least one Client has connected and you are now “live.” A Session Participants field shows you who is currently connected as a Client.



The DaVinci Remote Monitor showing the time connected and the participants involved

To Connect to a DaVinci Remote Monitor session as a Resolve Client:

- 1 The Resolve Client needs to have the full DaVinci Resolve Studio version of the software installed, or the iOS app installed.
- 2 Launch the DaVinci Remote Monitor app on your device. This is found in your DaVinci Resolve folder on your computer where you installed the software.
- 3 Sign into your Blackmagic Cloud account.
- 4 Paste the Session Code you received from the Resolve Host into the provided field.
- 5 Select the Output Device you wish to use for monitoring. This can be a connected computer display, or a Blackmagic Decklink or UltraStudio device for color accurate monitoring.
- 6 Click the Join button.



The DaVinci Remote Monitor app Join screen

At this point your Remote Monitor window will appear, and you will see and hear the output of the current viewer of the Resolve Host as they work.

NOTE: Remote Monitoring uses WebRTC to initiate connections between DaVinci Resolve Studio and multiple clients. However, some heavily restricted network setups may still block the initiation of a session, and a VPN may be required. A wired connection is recommended for the best performance.

To End a DaVinci Remote Monitor session as the Resolve Host:

- 1 Click on the DaVinci Remote Monitor icon in the lower right of the interface.
- 2 Click on the End Session button to stop the Remote Monitor for all users, or click on the X next to the name of a Client to disconnect the session for that user only.

To Leave a DaVinci Remote Monitor session as a Resolve Client:

- 1 Make the DaVinci Remote Monitor app active (not the Remote Monitor window).
- 2 Click on the Leave button to end your session.

To Change the Codec and Bitrate of a current DaVinci Remote Monitor session as the Resolve Host:

- 1 Click on the DaVinci Remote Monitor icon in the lower right of the interface.
- 2 Click on the Settings icon in the upper right of the Remote Monitor window.
- 3 Select a new Codec and Bitrate for the video stream.
- 4 Click on the Save button.

This will change the streaming settings to the new codec and bitrate that you chose. Existing Clients will be automatically reconnected to the new stream without having to do anything on their end. This lets you easily change the quality of the video stream on the fly to compensate for bandwidth and hardware issues.

DaVinci Remote Monitoring Using IP Address Connections

As an option, you can use Remote Monitoring over normal IP Address Connections, rather than having to sign in through Blackmagic Cloud. However, you will need some technical knowledge on setting up your network appropriately.

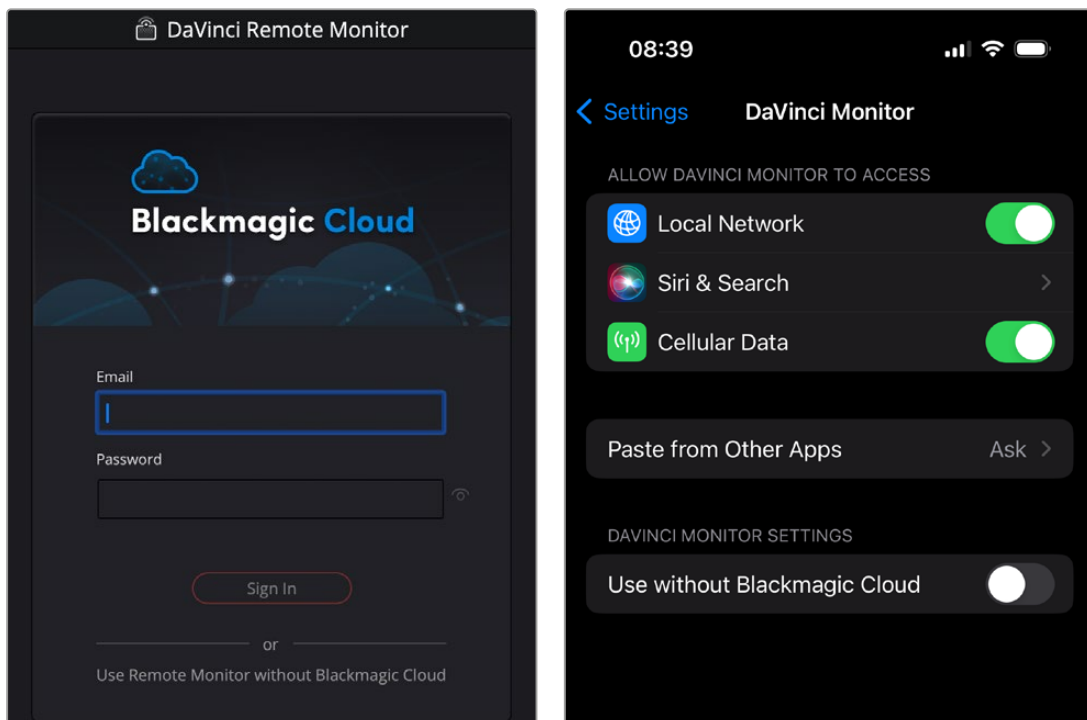
Setting up Remote Monitoring over IP connections on the Resolve Host

Follow the instructions below to set up your host grading computer for other people to monitor.

- 1 Make sure you are signed out of Blackmagic Cloud in Preferences > Internet Accounts.
- 2 In Preferences > System > General, check the "Use Remote Monitoring without Blackmagic Cloud" box.
- 3 Select Workspace > Remote Monitoring.
- 4 Adjust the codec and bit-rate parameters you want, then click Start Session.
- 5 Share the IP Address or configured host name for the Resolve host with the clients.

Setting up Remote Monitoring over IP connections on the Resolve Clients

Follow the instructions below to set up your client computer to monitor the host's workstation.



The Use Remote Monitor without Blackmagic Cloud settings in the computer and iOS versions

- 1 Launch the DaVinci Remote Monitor application.
- 2 In the Remote Monitor setup dialog, choose “Use Remote Monitor without Blackmagic Cloud.”
- 3 In the iOS or iPadOS versions of the App, go to the iOS Settings > DaVinci Monitor, and toggle on “Use Remote Monitor without Blackmagic Cloud.”
- 4 Enter the Resolve host's server IP Address or Host Name in the Host field.
- 5 Click Join.

Firewall exceptions and appropriate port forwarding will need to be manually configured in order to use this mode. The procedure to initiate a session and accept connections remains the same. This mode uses TCP server port 16410 and TCP/UDP port 16411, 16412, etc. for each client.

DaVinci Remote Monitoring via User Specified TURN Servers

When collaborating across complex network security setups that restrict the WebRTC STUN protocol, you may now decide to use self-configured or cloud-based TURN servers to route and relay monitoring streams.

To use this function, check the box under Preferences > System > General > Use TURN server for Remote Monitoring. Then add the URL of your custom TURN server.

DaVinci Remote Monitor Restrictions

There are currently some limitations to the DaVinci Remote Monitor application to be aware of.

- When connecting over the internet, bandwidth restrictions can hamper performance. If the bandwidth drops too low or cuts out completely, the Host will disconnect with an error message.
- Currently DaVinci Remote Monitor only works in the Media, Cut, Edit, Color, and Deliver pages.
- Currently Audio is limited to only 2 channels.



Advanced Workflows

CONTENTS

199	Workflow Integrations	4203
200	Creating DCTL LUTs.....	4210
201	TCP Protocol for DaVinci Resolve Transport Control.....	4215